```
\newcommand{\vect}{(x_1,x_2,\dots,x_n)}
```

Then we can type `$\vect$` anywhere after wards to produce $(x_1, x_2, \dots, x_n)$ as in

```
We often write $x$ to denote the vector $\vect$.
```

to get

> We often write x to denote the vector $(x_1, x_2, \dots, x_n)$.

(By the way, the best place to keep such "newcommands" is the preamble, so that you can use them anywhere in the document. Also, it will be easier to change the commands, if the need arises).

OK, we can now produce $(x_1, x_2, \dots, x_n)$ with `$\vect$`, but how about $(y_1, y_2, \dots, y_n)$ or $(z_1, z_2, \dots, z_n)$? Do we have to define newcommands for each of these? Not at all. We can also define commands with *variable arguments* also. Thus if we change our definition of `\vect` to

```
\newcommand{\vect}[1]{(#1_1,#1_2,\dots,#1_n)}
```

Then we can use `$\vect{x}$` to produce $(x_1, x_2, \dots, x_n)$ and `$\vect{a}$` to produce $(a_1, a_2, \dots, a_n)$ and so on.

The form of this definition calls for some comments. The `[1]` in the `\newcommand` above indicates that the command is to have *one* (variable) argument. What about the `#1`? Before producing the output, each occurrence of `#1` will be replaced by the (single) argument we supply to `\vect` in the input. For example, the input `$\vect{a}$` will be changed to `$(a_1,a_2,\dots,a_n)$` at some stage of the compilation.

We can also define commands with more than one argument (the maximum number is 9). Thus for example, if the document contains not only $(x_1, x_2, \dots, x_n)$, $(y_1, y_2, \dots, y_n)$ and so on, but $(x_1, x_2, \dots, x_m)$, $(y_1, y_2, \dots, y_p)$ also, then we can change our definition of `\vect` to

```
\newcommand{\vect}[2]{(#1_1,#1_2,\dotsc,#1_#2)}
```

so that we can use `$\vect{x}{n}$` to produce $(x_1, x_2, \dots, x_n)$ and `$\vect{a}{p}$` to produce $(a_1, a_2, \dots, a_p)$.

VIII.3. MORE ON MATHEMATICS

There are some many other features of typesetting math in LaTeX, but these have better implementations in the package amsmath which has some additional features as well. So, for the rest of the chapter the discussion will be with reference to this package and some allied ones. Thus all discussion below is under the assumption that the package amsmath has been loaded with the command `\usepackage{amsmath}`.

VIII.3.1. Single equations

In addition to the LaTeX commands for displaying math as discussed earlier, the amsmath also provides the `\begin{equation*} ... \end{equation*}` construct. Thus with this package loaded, the output

> The equation representing a straight line in the Cartesian plane is of the form
>
> $$ax + by + c = 0$$
>
> where a, b, c are constants.

can also be produced by

```
The equation representing a straight line in the Cartesian plane is
of the form
\begin{equation*}
  ax+by+c=0
\end{equation*}
where $a$, $b$, $c$ are constants.
```

Why the * after `equation`? Suppose we try it without the * as

```
The equation representing a straight line in the Cartesian plane is
of the form
\begin{equation}
  ax+by+c=0
\end{equation}
where $a$, $b$, $c$ are constants.
```

we get

The equation representing a straight line in the Cartesian plane is of the form

$$ax + by + c = 0 \tag{VIII.1}$$

where a, b, c are constants.

This provides the equation with a *number*. We will discuss equation numbering in some more detail later on. For the time being, we just note that for any environment name with a star we discuss here, the unstarred version provides the output with numbers.

Ordinary text can be inserted inside an equation using the `\text` command. Thus we can get

Thus for all real numbers x we have

$$x \le |x| \quad \text{and} \quad x \ge |x|$$

and so

$$x \le |x| \quad \text{for all } x \text{ in } R.$$

from

```
Thus for all real numbers $x$ we have
\begin{equation*}
  x\le|x|\quad\text{and}\quad x\ge|x|
\end{equation*}
and so
\begin{equation*}
  x\le|x|\quad\text{for all $x$ in $R$}.
\end{equation*}
```

Note the use of dollar signs in the second `\text` above to produce mathematical symbols within `\text`.

Sometimes a single equation maybe too long to fit into one line (or sometimes even *two* lines). Look at the one below:

$$(a+b+c+d+e)^2 = a^2+b^2+c^2+d^2+e^2 \\ + 2ab+2ac+2ad+2ae+2bc+2bd+2be+2cd+2ce+2de$$

This is produced by the environment `multline*` (note the spelling carefully—it is *not* `mult i line`), as shown below.

```
\begin{multline*}
  (a+b+c+d+e)^2=a^2+b^2+c^2+d^2+e^2\\
       +2ab+2ac+2ad+2ae+2bc+2bd+2be+2cd+2ce+2de
\end{multline*}
```

`multline` can be used for equations requiring more than two lines, but without tweaking, the results are not very satisfactory. For example, the input

```
\begin{multline*}
  (a+b+c+d+e+f)^2=a^2+b^2+c^2+d^2+e^2+f^2\\
       +2ab+2ac+2ad+2ae+2af\\
       +2bc+2bd+2be+2bf\\
       +2cd+2ce+2cf\\
       +2de+2df\\
       +2ef
\end{multline*}
```

produces

$$(a+b+c+d+e+f)^2 = a^2+b^2+c^2+d^2+e^2+f^2 + 2ab+2ac+2ad+2ae+2af + 2bc+2bd+2be+2bf + 2cd+2ce+2cf + 2de+2df + 2ef$$

By default, the `multline` environment places the first line flush left, the last line flush right (except for some indentation) and the lines in between, centered within the display.

A better way to typeset the above multiline (not multline) equation is as follows.

$$\begin{aligned}(a+b+c+d+e+f)^2 &= a^2+b^2+c^2+d^2+e^2+f^2\\ &\quad +2ab+2ac+2ad+2ae+2af\\ &\quad +2bc+2bd+2be+2bf\\ &\quad +2cd+2ce+2cf\\ &\quad +2de+2df\\ &\quad +2ef\end{aligned}$$

This is done using the `split` environment as shown below.

```
\begin{equation*}
  \begin{split}
    (a+b+c+d+e+f)^2 & = a^2+b^2+c^2+d^2+e^2+f^2\\
                    &\quad +2ab+2ac+2ad+2ae+2af\\
                    &\quad +2bc+2bd+2be+2bf\\
                    &\quad +2cd+2ce+2cf\\
                    &\quad +2de+2df\\
                    &\quad +2ef
  \end{split}
\end{equation*}
```

Some comments seems to be in order. First note that the `split` environment cannot be used independently, but only inside some equation structure such as `equation` (and others we will soon see). Unlike `multline`, the `split` environment provides for alignment among the "split" lines (using the & character, as in `tabular`). Thus in the above example, all the + signs are aligned and these in turn are aligned with a point a `\quad` to the right of the = sign. It is also useful when the equation contains multiple equalities as in

$$\begin{aligned}(a+b)^2 &= (a+b)(a+b)\\ &= a^2+ab+ba+b^2\\ &= a^2+2ab+b^2\end{aligned}$$

which is produced by

```
\begin{equation*}
  \begin{split}
    (a+b)^2 & = (a+b)(a+b)\\
            & = a^2+ab+ba+b^2\\
            & = a^2+2ab+b^2
  \end{split}
\end{equation*}
```

VIII.3.2. Groups of equations

A group of displayed equations can be typeset in a single go using the `gather` environment. For example,

$$\begin{aligned}(a,b)+(c,d) &= (a+c,b+d)\\ (a,b)(c,d) &= (ac-bd,ad+bc)\end{aligned}$$

can be produced by

```
\begin{gather*}
  (a,b)+(c,d)=(a+c,b+d)\\
  (a,b)(c,d)=(ac-bd,ad+bc)
\end{gather*}
```

Now when several equations are to be considered one unit, the logically correct way of typesetting them is with some alignment (and it is perhaps easier on the eye too). For example,

Thus x, y and z satisfy the equations

$$\begin{aligned}x+y-z &= 1\\ x-y+z &= 1\end{aligned}$$

This is obtained by using the `align*` environment as shown below

```
Thus $x$, $y$ and $z$ satisfy the equations
\begin{align*}
  x+y-z & = 1\\
  x-y+z & = 1
\end{align*}
```

We can add a short piece of text between the equations, without disturbing the alignment, using the `\intertext` command. For example, the output

Thus x, y and z satisfy the equations

$$x+y-z = 1$$
$$x-y+z = 1$$

and by hypothesis

$$x+y+z = 1$$

is produced by

```
Thus $x$, $y$ and $z$ satisfy the equations
\begin{align*}
  x+y-z & = 1\\
  x-y+z & = 1\\
  \intertext{and by hypothesis}
  x+y+z & =1
\end{align*}
```

We can also set multiple 'columns' of aligned equations side by side as in

Compare the following sets of equations

$$\cos^2 x + \sin^2 x = 1 \qquad \cosh^2 x - \sinh^2 x = 1$$
$$\cos^2 x - \sin^2 x = \cos 2x \qquad \cosh^2 x + \sinh^2 x = \cosh 2x$$

All that it needs are extra `&`'s to separate the columns as can be sen from the input

```
Compare the following sets of equations
\begin{align*}
  \cos^2x+\sin^2x & = 1        & \cosh^2x-\sinh^2x & = 1\\
  \cos^2x-\sin^2x & = \cos 2x & \cosh^2x+\sinh^2x & = \cosh 2x
\end{align*}
```

We can also adjust the horizontal space between the equation columns. For example,

```
Compare the sets of equations
\begin{align*}
  \cos^2x+\sin^2x & = 1        &\qquad \cosh^2x-\sinh^2x & = 1\\
  \cos^2x-\sin^2x & = \cos 2x &\qquad \cosh^2x+\sinh^2x & = \cosh 2x
\end{align*}
```

gives

Compare the sets of equations

$$\cos^2 x + \sin^2 x = 1 \qquad \cosh^2 x - \sinh^2 x = 1$$
$$\cos^2 x - \sin^2 x = \cos 2x \qquad \cosh^2 x + \sinh^2 x = \cosh 2x$$

Perhaps a nicer way of typesetting the above is

Compare the following sets of equations

$$\begin{aligned} \cos^2 x + sin^2 x &= 1 \\ \cos^2 x - \sin^2 x &= \cos 2x \end{aligned} \quad \text{and} \quad \begin{aligned} \cosh^2 x - \sinh^2 x &= 1 \\ \cosh^2 x + \sinh^2 x &= \cosh 2x \end{aligned}$$

This cannot be produced by the equation structures discussed so far, because any of these environments takes up the entire width of the text for its display, so that we cannot put anything else on the same line. So amsmath provides variants `gathered`, `aligned` and `alignedat` which take up only the *actual width of the contents* for their display. Thus the above example is produced by the input

```
Compare the following sets of equations
\begin{equation*}
  \begin{aligned}
    \cos^2x+sin^2x  & = 1\\
    \cos^2x-\sin^2x & = \cos 2x
  \end{aligned}
  \qquad\text{and}\qquad
  \begin{aligned}
    \cosh^2x-\sinh^2x & = 1\\
    \cosh^2x+\sinh^2x & = \cosh 2x
  \end{aligned}
\end{equation*}
```

Another often recurring structure in mathematics is a display like this

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x \leq 0 \end{cases}$$

There is a special environment `cases` in amsmath to take care of these. The above example is in fact produced by

```
\begin{equation*}
  |x| =
  \begin{cases}
    x & \text{if $x\ge 0$}\\
    -x & \text{if $x\le 0$}
  \end{cases}
\end{equation*}
```

VIII.3.3. Numbered equations

We have mentioned that each of the the 'starred' equation environments has a corresponding unstarred version, which also produces numbers for their displays. Thus our very first example of displayed equations with `equation` instead of `equation*` as in

```
The equation representing a straight line in the Cartesian plane is
of the form
\begin{equation}
  ax+by+c=0
\end{equation}
where $a$, $b$, $c$ are constants.
```

produces

The equation representing a straight line in the Cartesian plane is of the form

$$ax + by + c = 0 \tag{VIII.2}$$

where a, b, c are constants.

Why VIII.2 for the equation number? Well, this is Equation number 2 of Chapter VIII, isn't it? If you want the section number also in the equation number, just give the command

```
\numberwithin{equation}{section}
```

We can also override the number LaTeX produces with one of our own design with the `\tag` command as in

```
The equation representing a straight line in the Cartesian plane is
of the form
 \begin{equation}
  ax+by+c=0\tag{L}
\end{equation}
where $a$, $b$, $c$ are constants.
```

which gives

The equation representing a straight line in the Cartesian plane is of the form

$$ax + by + c = 0 \tag{L}$$

where a, b, c are constants.

There is also a `\tag*` command which typesets the *equation label* without parentheses.

What about numbering alignment structures? Except for `split` and `aligned`, all other alignment structures have unstarred forms which attach numbers to *each* aligned equation. For example,

```
\begin{align}
  x+y-z &= 1\\
  x-y+z &= 1
\end{align}
```

gives

$$x + y - z = 1 \tag{VIII.3}$$
$$x - y + z = 1 \tag{VIII.4}$$

Here is also, you can give a label of your own to *any* of the equations with the `\tag` command. Be careful to give the `\tag` *before* the end of line character `\\` though. (See what happens if you give a `\tag` command *after* a `\\`.) You can also suppress the label for any equation with the `\notag` command. These are illustrated in the sample input below:

```
Thus $x$, $y$ and $z$ satisfy the equations
\begin{align*}
```